Paper Title	Year	Input & Output	Text Encoder	Vision Encoder	Multimodal Fusion	Decoder / LLM	Training Method	Datasets	Arabic Dialects	Results	Limitation	Added Value
Dallah: A Dialect-Aware Multimodal Large Language Model for Arabic	2024	In: Image + text Out: Arabic text response in matching dialect or MSA	AraLLaMA Tokenizer & Embedding module	CLIP ViT-L/14 (336×336 px, 576 tokens)	2-layer MLP connector	AraLLaMA-7B (Arabic-adapted LLaMA-2)	3-stage: (1) Projector PT on LCS-558K (ViT/LLM frozen); (2) LoRA SFT on LLaVA-Instruct; (3) Dialect SFT across 6 dialects.	Pretrain: LCS-558K (800k AR+EN); SFT: LLaVA-Instruct (139k AR); Dialect: ~4k QA pairs (EG, MR, MA, PS, SA, YE).	6 dialects (Egyptian, Mauritanian, Moroccan, Palestinian, Saudi, Yemeni) + MSA.	Ara-LLaVA-Bench: 89.7% (GPT-4), 74.6% (Human); beats PALO (+1.7%) and Peacock (+10.8%). Dallah-Bench: Human DA 6.87 / CA 7.09.	Identifies Western figures but misses Arab cultural icons; Dialect mixing; no scene-text OCR.	First 6-dialect aware Arabic MLLM; embedding-based back-translation filter; released Dallah-Bench benchmark.
Peacock: A Family of Arabic Multimodal Large Language Models and Benchmarks	2024	In: Image + Arabic prompt Out: Arabic text response	LLM Tokenizer & Embeddings; BERT-base Q-Former for visual query extraction	InstructBLIP: ViT-G/14 (224px); LLaVA: CLIP ViT-L/14 (336px)	InstructBLIP: Q-Former + Linear layer; LLaVA: 2-layer MLP	AraLLaMA-7B or AceGPT-7B-chat	2-stage: (1) Projector PT on LCS-558K + COCO (ViT/LLM frozen); (2) LoRA SFT on LLaVA-Instruct, VQAv2, MCQs.	LCS-558K (389k), COCO (527k), LLaVA-Instruct (204k), VQAv2 (353k), OKVQA (7k), GQA (638k); filtered via LaBSE.	Primary focus on MSA; 1k exploratory pair case study on Egyptian dialect.	InstructBLIP-AraLLaMA: VQAv2 56.2% EM, OKVQA 29.8%, GQA 58.1%, LLaVA-Bench 82.3 avg; Henna: 62.3 Helpfulness / 69.0 Relevance.	Object hallucination; MT polysemy errors (e.g. 'sitting' as 'lying down'); lacks Arabic scene-text OCR.	Comprehensive suite comparing Q-Former vs. MLP; introduced AraLLaMA-7B; LaBSE filter (280%); Henna benchmark.
AIN: The Arabic INclusive Large Multimodal Model	2025	In: Images, video, or documents + EN/AR query Out: Arabic or English text response (MSA/EN)	Qwen2-VL native Tokenizer & Embeddings	Qwen2-VL native Vision Transformer (ViT)	Qwen2-VL native cross-attention / projection blocks	Qwen2-VL-7B (full-parameter fine-tuned)	Full-parameter fine-tuning for 1 epoch on 64× A100 (80GB) GPUs using FlashAttention-2 and LLaMA-Factory.	3.6M instruction pairs across 8 domains (35% authentic Arabic); translated via GPT-4o-mini + LaBSE filter (≥80%).	Modern Standard Arabic (MSA).	CAMEL-Bench: 63.8 (beats GPT-4o 60.1, Qwen2-VL 52.6); ArabicMMLU: 59.4; Human survey: 76% preferred AIN over GPT-4o (15%).	MSA focus.	First comprehensive bilingual EN-AR MLLM across 8 specialized domains; SOTA on Arabic OCR and document understanding.
CultureVLM: Characterizing and Improving Cultural Understanding of Vision-Language Models for over 100 Countries	2025	In: Image + 4-option cultural MCQ in English Out: Selected answer letter + reasoning	Inherited from base models (LLaVA-1.5, Phi-3-Vision, LLaMA-3.2-Vision)	Inherited from base models (CLIP ViT-L/14, SigLIP)	Inherited linear / MLP connectors	Fine-tuned LLaVA-1.5 (7B/13B), Phi-3-Vision (4B), and LLaMA-3.2-Vision (11B)	Supervised fine-tuning on CultureVerse.	CultureVerse: 19,682 concepts across 188 countries, 74,959 images, 196,673 train QA pairs, 31,382 test QAs.	Evaluated culturally across Middle East and North Africa (MENA) regions; questions in English.	LLaVA-7B cultural knowledge surged from 48.1% to 90.9%; general reasoning preserved (ScienceQA 70.1% → 69.3%); strong cross-region transfer.	Stepwise reasoning prompting hurts visual recognition; does not evaluate in native Arabic dialects; reliant on English framing.	First global cultural VLM benchmark (19k concepts, 188 nations); proved small-data cultural fine-tuning effectively enhances cultural awareness.
PhantomTroupe at ImageEval 2025 Shared Task: Multimodal Arabic Image Captioning through Translation-Based Fine-Tuning of LLM Models	2025	In: Image (grayscale) + Arabic prompt Out: Arabic caption	Qwen2.5-VL native Tokenizer & Embeddings	Built-in Qwen2.5-VL Vision Transformer (ViT)	Native Qwen2.5-VL cross-attention / projection blocks	Qwen2.5-VL-7B-Instruct (4-bit NF4 quantized via BitsAndBytes)	PEFT via LoRA (r=64, α=64) on vision & text attention heads; bidirectional translation (AR → EN → AR) augmentation; grayscale conversion.	ImageEval 2025 Subtask 2: 3,071 annotated Palestinian culture images (2,717 train, 75 val, 279 blind test).	MSA with Levantine/Palestinian cultural context.	Ranked 3rd overall (BLEU-1: 22.6, Cosine Similarity: 57.5, Judge: 31.4), significantly outperforming LLaMA-3.2-11B and Pixtral-12B baselines.	Errors propagate from intermediate MT; defaults to MSA over regional vernaculars; small dataset (3k images); constrained compute.	Parameter-efficient 4-bit LoRA fine-tuning recipe for culturally sensitive Arabic captioning; bidirectional translation augmentation pipeline.